

Wrap-Up of Virtuous Data Science

DATA 202 21FA

Today

- Please fill out *Course evaluations*
 - It's helpful *data* for me, my department, and the university.
 - Each comment matters. Each rating matters.
 - Be honest, be balanced.
 - Come get me in the hall when everyone is done.
- Virtuous Data Science: wrap-up

Logistics

Sign up for final project presentation slot.

Q&A

| Is "report" just the knit PDF?

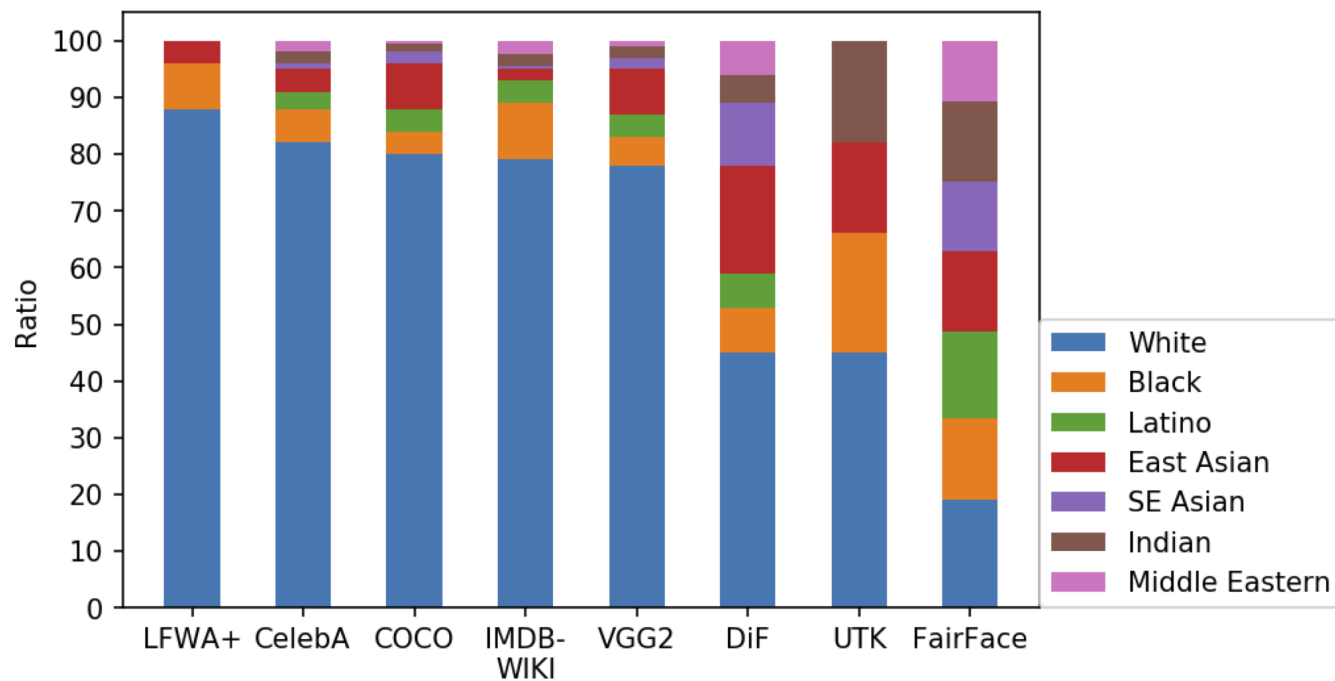
Yes. But make sure it's *readable* as a *standalone document* by someone who *can't read R code*.

| Is data visualization a separate job?

Sometimes, e.g., NYTimes Graphics Dept (@nytgraphics). But data scientists often wear many hats.

Q&A

Why were facial recognition systems less accurate for minorities?





Are there ways that we can make this better for the future?

Measurement. Accountability. Diverse representation on team leadership. ...

...some of your quotes...

Recommendation Systems

people begin to try to fit into a box

Part of me wants to find this creepy, but at the same time its somewhat convenient

... mindlessly scroll through the platform's content without thinking very much ... However, the platforms ... are filtering what I am seeing

Facial Recognition

I think as a society we have become obsessed with the advantages of automation and maybe have not taken the proper time to consider the disadvantages automation causes.

When we fail to acknowledge diversity we create systems that disproportionately affect others.

Wrap-Up of Virtuous Data Science Objectives

- Describe how Reformed concepts of justice and shalom apply to data collection, analysis, sharing, and use.
- Give examples of specific concerns around privacy, bias, accountability, and transparency
- Describe steps and dispositions that individual data scientists can take to act justly in their profession

A bullet-point summary of biblical *justice*

- Community above individual (voluntarily)
- Equity: equal treatment, dignity
- Collective responsibility
- Individual responsibility
- Advocacy for poor and marginalized

What does biblical justice require, in the area of data science?

Community

The righteous are willing to disadvantage themselves to advantage the community; the wicked are willing to disadvantage the community to advantage themselves.

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- privacy: what *must* we share? what must we *not* share?
- integrity in data collection, analysis, reporting, communication

Thus:

- Data analysis process: *reproducible, transparent, documented*
- Reporting: transparency about *limitations, choices*, consideration of *possible harms*

Equity: Everyone must be treated equally and with dignity.

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- *direct* impact
 - fair risk assessment (see Discussion and COMPAS)
 - fair resource allocation
 - fair surveillance (don't hyper-surveil the poor etc.)
- *indirect* impact:
 - don't show ads for criminal background checks more often for Black names
 - don't tolerate higher speech reco error rates for minorities
 - show a representative diversity of age/gender/race/... in image searches

Should we even be predicting peoples' lives?

- Risk assessment for criminality, loan approval, etc. requires predicting peoples' future actions and situations
- These predictions might be terribly inaccurate. *Should we be trying at all?*

Salganik et al, **Measuring the predictability of life outcomes with a scientific mass collaboration**. PNAS, April 2020

Despite using a rich dataset and applying machine-learning methods optimized for prediction, the best predictions were not very accurate and were only slightly better than those from a simple benchmark model.

Corporate responsibility: I am sometimes responsible for and involved in other people's sins.

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- Even if I intend no prejudice, my *algorithm* could be prejudiced because of training data.
- Even if my work is honest, I could be supporting a company that exploits other workers directly or rely on conflict minerals and **child labor**
- Environmental responsibility is both individual and collective

Individual responsibility: I am finally responsible for all my sins, but not for all my outcomes.

Individual responsibility: I am finally responsible for all my sins, but not for all my outcomes.

- I must do what's right, whether or not my company's policies require it.
- When something isn't right, I need to say something even if it risks my job.

Advocacy: We must have special concern for the poor and the marginalized.

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- By **listening to** and **amplifying**, not **speaking for**.
- e.g., beware of doing "parachute research" or de-contextualized "Data for Good"

Data science, and data scientists, *are not saviors*.

Incarnation

In your relationships with one another, have the same mindset as Christ Jesus:

Who, being in very nature God,
did not consider equality with God something
to be used to his own advantage;
rather, he made himself nothing
by taking the very nature of a servant,
being made in human likeness.

And being found in appearance as a man,
he humbled himself
by becoming obedient to death—
even death on a cross!

Learning More

Some further reading on data ethics

- The Oxford Handbook of Ethics of AI
- Coded Bias documentary
- Fast.AI Data Ethics course
- **Ethics and Data Science** by Mike Loukides, Hilary Mason, DJ Patil
- **Weapons of Math Destruction: *How Big Data Increases Inequality and Threatens Democracy***, by Cathy O'Neil
- **How Charts Lie: *Getting Smarter about Visual Information***, by Alberto Cairo
- **How Deceptive are Deceptive Visualizations?** Pandey et al., CHI 2015

Other resources on Data Ethics

- AI Now Institute
- Data and Society
- AlgorithmWatch
- Harvard BKC
- ACM Conference on Fairness, Accountability, and Transparency (FAccT)

Who/What I'm Reading / Following: Tech

- RStudio AI blog
- tidyverse blog
- RWeekly
- distill.pub
- Harvard Data Science Review
- TWiML Podcast
- Cassie Kozyrkov (@quaesita)

"What can I do?"

- Practice the "data dispositions"
 - Humility (cite sources, acknowledge limitations, validate results)
 - Integrity (check assumptions, reproduce analyses, evaluate others' claims)
 - Hospitality (clear visuals, clear reports, clear code)
 - Compassion and justice
- Listen a lot. To diverse opinions. (e.g., "The Flip Side")
- Keep in touch.